What the EEG-to-BOLD evidence supports

Resting EEG can support prediction research. Faithful regional reconstruction requires more than temporal correlation.

Existing ridge predictions: mean raw R^2 0.094 across 64 regions.

EEG predicts better than these physiology features.

The next architecture change is a controlled hypothesis, not a demonstrated improvement.

What NeuroBOLT computes

Waveform and spectral encoders summarize the same EEG history; the model adds their pooled representations.

The completed comparison still favours ridge

Historical five-fold temporal r: ridge .381; shared NeuroBOLT 24 s .336; shared NeuroBOLT 16 s .323.

Correlation is not waveform accuracy

The earlier physiology partition used signed squared correlation under an R^2 label. The correction uses prediction error.

People are the units of uncertainty

EEG: R^2 0.094 ± 0.094; physiology 0.044 ± 0.049; joint 0.092 ± 0.096.

EEG minus physiology: ΔR^2 +0.0497; 95% CI [0.0185, 0.0828]; p=0.0063; BY=0.0189; d_z=0.639; n=22.

Global-signal regression defines another target

For existing residual-trained ridge, mean regional r=0.144, raw R^2=0.010. GSR is retrospective.

Lag patterns need participant uncertainty

Alpha peak difference, thalamus minus cortex: -6 s; bootstrap interval [-8, -4] s. Resolution: 2 s.

The checkpoint gate limits neural claims

CKA/IG were blocked by missing CV checkpoints. Existing waveform swaps below remain descriptive evidence from five people.

Next test: physical spectral-lag alignment

Replace one dense time-axis mapping in spectral-only NeuroBOLT24 with physical support-overlap weights.

A two-week investigation with a stopping rule

Complete one controlled change and explain its result. Expand only after a profiled comparison and a new compute budget.